



Nano Project Initiative for Undergraduate Research and Development Fellow (UGRDF)

Instructions for Completing the Assignment:

1. **Choose a Topic:** Select a topic that interests you and is feasible to work on within the given timeline.
2. **Identify a Guide:** Find a guide or mentor who has expertise in the selected topic.
3. **Discuss and Finalize the Problem Statement:**
 - 0 Schedule a discussion with your guide.
 - Define a small, manageable problem statement that can be addressed within 7 days.
 - Finalize the problem statement during the discussion and begin your work immediately.
4. **Conduct a Literature Survey:**
 - 0 Study and analyze at least 10 relevant research papers.
 - Include at least one patent related to the topic's application.
 - Review 5 relevant news articles.
 - Use a reference manager to organize and cite all these resources properly, as these will be referenced in your report.
5. **Develop a Methodology:**
 - 0 Outline your approach to the literature survey, problem-solving, and the work planned for the following days.
 - Document this methodology clearly in your report.
6. **Execute the Methodology and Record Results:**
 - 0 Implement the planned steps, following your methodology.
 - Document the results of your work, whether they meet expectations or not.
 - Remember, this is a short-term assignment, not a groundbreaking research project. Keep it straightforward but substantial enough to demonstrate your effort.
7. **Write a Discussion Section:**
 - 0 Analyze your results and present your understanding of the topic and findings in detail.
8. **Prepare the Final Report:**
 - 0 Structure your report formally with the following sections:
 - Introduction
 - Literature Review
 - Methodology
 - Results
 - Discussion
 - Conclusion
 - References (ensure citations meet the minimum required as outlined above) ○
 - Include citations wherever necessary.
9. **Engage with Your Guide:**
 - 0 Keep your guide informed throughout the process.
 - Ensure your guide is satisfied with your work and provides constructive feedback.

10. **Obtain a Recommendation:**

- 0 At the end of your assignment, request your guide to complete the recommendation form, which will be provided online.

11. **Submission and Evaluation:**

- 0 Submit your completed report, ensuring all appendices are included before the table of contents, with each appendix on a separate page.
- o Note: Appendices 8 and 9, provided for reference, will be shared online and do not need to be attached.
- o Submit your completed work as a ppt.
- o Your report and PPT will be reviewed based on the quality of your work and guide feedback.
- o Successful candidates will be inducted into the research cell as **UG Research and Development Fellows**.

Final Note:

You are embarking on an exciting journey, and while it may seem challenging, remember that you are not alone. Here are some key tips and advice to help you succeed and maintain your well-being:

1. **Seek Help When Needed:**

- 0 Don't hesitate to ask for technical assistance or emotional support.
- o Speak to your seniors, peers, or mentors if you encounter difficulties.

2. **Maintain a Healthy Work Environment:**

- 0 Work in a properly ventilated area to stay focused and comfortable.
- o Use resources like the library, R&D labs, or dedicated study halls to optimize productivity.

3. **Prioritize Your Well-Being:**

- 0 Ensure your physical and mental health comes first.
- o Take adequate sleep, eat nutritious meals, and stay hydrated.

4. **Take Breaks to Recharge:**

- 0 Continuous work without rest can reduce productivity.
- o Remember, a knife cuts better after being sharpened—take breaks to rejuvenate yourself.

5. **Communicate Effectively:**

- 0 If you feel your chosen guide is not the right fit, let us know without hesitation.
- o We are here to support you and will help find a suitable solution while ensuring your academic commitments are unaffected.

6. **Stay Balanced:**

- 0 Avoid burning out by setting realistic goals and managing your time effectively.
- o Focus on the quality of your work rather than overwhelming yourself with excessive tasks.

7. **Use Available Resources:**

- 0 Take full advantage of the tools and facilities provided to you.

-
- Collaborate with others to gain insights and broaden your perspective.
8. **Stay Positive and Motivated:**
 - Challenges are part of the process; use them as learning opportunities.
 - Celebrate small victories along the way to keep your morale high.
 9. **Ask for Guidance Anytime:**
 - Whether it's technical queries, mental support, or logistical assistance, we are here to help you every step of the way.
 - Your success and satisfaction are our top priorities, and we'll handle your concerns with the utmost care.
 10. **Reflect and Refine:**
 - Regularly evaluate your progress and adjust your methodology if needed.
 - Learn from any mistakes and focus on continuous improvement.

We believe in your potential to achieve great results. Stay determined, take care of yourself, and give it your best. All the best for this exciting endeavor!

Appendix 1: First Page Template (Don't use this line, copy from below)

Kumaraguru College of Technology

Nano project for the partial fulfillment

of selection for

Undergraduate Research and Development Fellow (UGRDF)

Name of the Candidate : THARUN N E
Department : ELECTRONICS AND INSTRUMENTATION
ENGINEERING
Title of the Assignment/Research : A Review on Grid Enhancement
Strategies Enabled by Solar Energy Inflows
Duration of the Assignment : FROM 23|11|2025 TO 1|12|2025

Guide's Name : DR.RANGANATHAN S
Guide's Designation : ASSISTANT PROFESSOR 2
Guide's Contact Information :

Signature :



Date : 1|12|2025

Appendix 2: Declaration Form (Plagiarism Check)

Declaration Form

I, THARUN N E (Student Name), hereby declare that:

1. The work presented in this project is my original work.
2. I have cited all sources used in the project properly.
3. I understand the institution's policy on academic integrity and plagiarism, and this work complies with the guidelines.

Signature : THARUN N E

Date : 1|12|2025

Appendix 3: Program Feedback Form (Student)

Feedback Form for Nano Project Initiative

Name THARUN N E

Role: Student

Rate the Following (1 – Poor, 5 – Excellent):

1. Clarity of Objectives: ☐ 1 ☐ 2 ☐ 3 ☒ 4 ☐ 5
2. Adequacy of Resources: ☐ 1 ☐ 2 ☐ 3 ☒ 4 ☐ 5
3. Quality of Guidance: ☐ 1 ☐ 2 ☐ 3 ☒ 4 ☐ 5
4. Overall Experience: ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☒ 5

Suggestions for Improvement:

I NEED A TECHNICAL HANDS ON PRACTICE . WHICH IS MY OPINION .

Signature THARUN N E

Date : 1|12|2025

Appendix 5: Progress Tracker/Work Log

Progress Tracker for Nano Project

Name THARUN N E **Project Title** : A Review on Grid Enhancement Strategies Enabled by Solar Energy Inflows

Date	Task Completed	Challenges Faced	Remarks by Guide
Day 1	RESEARCH PAPER	CANT FIND SUITABLE PAPER	GOOD
Day 2	FACULTY	NO PROBLEM FACED	GOOD
Day 3	DIFFICULTY IN FINDING PAPER	TOOK HOURS TO ANALYZE	GOOD
Day 4	FINDING FORMAT	RESEARCHED MANY PAPER FOR FORMAT	GOOD
Day 5	STARTED RESEARCH	LATE SLEEP	GOOD
Day 6	STARTED WRITING	PREPARED NOTES AND WROTE	GOOD
Day 7	COMPLETED	COMPLETED	GOOD

Student's Signature : THARUN N E

Guide's Signature : 

Appendix 6: Final Report

Template for Final Report

Title Page

- Title of the Project
 - Name of the Student
 - Guide's Name and Designation
 - Department and Institution
 - Submission Date
- Report Structure:**

1. **Introduction:** Define the problem, objectives, and significance of the project.
2. **Literature Survey:** Discuss the reviewed research papers, patents, and articles.
3. **Methodology:** Outline the step-by-step approach used in the project.
4. **Results:** Present findings, observations, or outputs (tables/graphs as needed).
5. **Discussion:** Analyze and interpret the results.

6. **Conclusion:** Summarize key insights and outcomes.
7. **References:** List all cited works in a standard format.

Appendix 7: Student Self-Assessment Form

Student Self-Assessment Form for Nano Project

Name : THARUN N E Project Title : **A Review on Grid Enhancement Strategies Enabled by Solar Energy Inflows**

Guide's Name : DR . RANGANATHAN V Duration
: From 24/11/2025 to 1/12/2025

1. **What did you learn during this project?**
2. During this research project, I learned how to locate credible sources, analyze information critically, and synthesize ideas into a clear, structured argument. I also gained a better understanding of my topic and improved my academic writing and time-management skills.
- 3.

What challenges did you face, and how did you overcome them?

One of the main challenges I faced was finding credible and up-to-date sources. At first, my searches were too broad, so I refined my keywords and focused on academic databases, which helped me gather more reliable information. Another challenge was organizing a large amount of data. I overcame this by creating an outline and grouping my notes into themes, which made writing the paper much more manageable.

3. Rate your performance (Self-Assessment):

- ☒ Excellent
- ☒ Good
- ☐ Satisfactory
- ☐ Needs Improvement

4. **Suggestions for improving the program:** already nice

Student's Signature : THARUN N E
Date : 1/12/2025

Appendix 8: Guide Feedback Form (Criteria)

Guide Feedback Form for Nano Project

Student's Name : THARUN N E_

Guide's Name : RENGANATHAN V

Project Title : **A Review on Grid Enhancement Strategies Enabled by Solar Energy Inflows**

Duration : 24/11/2025 TO 1/12/2025

Evaluation Criteria

1. Effort and Commitment:

- ☐ Excellent
- ☒ Good
- ☐ Satisfactory
- ☐ Needs Improvement

2. Understanding of the Problem Statement:

- ☐ Excellent
- ☒ Good
- ☐ Satisfactory
- ☐ Needs Improvement

3. Research and Problem-Solving Skills:

- ☐ Excellent
- ☒ Good
- ☐ Satisfactory
- ☐ Needs Improvement

4. Quality of Documentation and Reporting:

- ☐ Excellent
- ☒ Good
- ☐ Satisfactory
- ☐ Needs Improvement

5. Overall Performance:

- ☐ Excellent
- ☒ Good
- ☐ Satisfactory
- ☐ Needs Improvement

Additional Comments (Optional):

Guide's Signature : _____

Date : _____

A Review on Grid Enhancement Strategies Enabled by Solar Energy Inflows

ABSTRACT:

The study looks at the current situation of solar energy adoption worldwide and in India. It emphasizes the growing importance of renewable energy due to rising temperatures and increasing power demands. India ranks seventh in solar energy production, contributing 9 GW, with installations across government projects, private initiatives, and household systems. The researchers used a structured questionnaire and collected primary data from 105 respondents to identify the main obstacles to solar adoption. They analyzed these challenges using the Relative Importance Index (RII). The findings reveal that cost-related issues, high equipment prices, and restrictive subsidy policies—limited to purchases from government vendors—majorly hinder progress. Further analysis indicates that battery performance is greatly affected by temperature. Higher temperatures can temporarily boost storage capacity, but they also shorten battery lifespan over time. Lithium batteries work well but are more sensitive to temperature changes and come with a higher price. In contrast, flooded and gel batteries are more cost-effective in hot regions. The study concludes that expanding access to subsidies, conducting regional research, and improving policy reforms are crucial for enhancing solar energy integration in India.

1.INTRODUCTION :

Photovoltaic (PV) solar grid integration is essential for building a sustainable energy future. It allows two-way power flow from distributed solar units. The process begins with solar modules, which are made up of linked solar cells that turn sunlight into direct current (DC) power. Several modules are usually grouped together in solar arrays to increase the total power output. The generated DC power is supplied to one or more inverters. The inverter's main job is to change DC power into alternating current (AC) power. AC is the standard type of electricity used in the grid and by most household devices. The inverter also ensures that the frequency and phase of the AC output match the grid, allowing for a smooth and stable connection. After the inversion, the AC power typically goes through a transformer. In smaller, decentralized setups like rooftop PV systems, a low-voltage transformer raises the voltage to meet the needs of the local distribution network. In larger solar plants, the voltage is increased to higher levels for long-distance transmission. The grid-tie point, often featuring a net energy meter, connects the solar installation to the utility grid. The net meter tracks both the electricity drawn from the grid and any extra energy sent back from the solar system. In more advanced configurations, especially those designed to manage the variability of solar power, a battery energy storage system (BESS) may also be included[1]. The effective use of solar energy on a large scale requires its smooth integration into the current electrical grid. Grid integration means linking solar power systems to the utility grid. It ensures the reliable and stable operation of the entire power system. This integration is crucial for making the most of solar energy. It allows solar power to significantly contribute to the national energy mix while reducing reliance on traditional fossil fuels. However, connecting a variable energy source like solar power to a grid designed for consistent electricity production presents unique challenges. This review paper aims to provide a detailed evaluation of these challenges. It will also look at the methods used or suggested in India to address them. Finally, it will discuss the future prospects for solar grid integration in the country, based on recent research and developments in this field[1]. Renewable energy systems like wind power plants and solar photovoltaic (PV) units are being installed quickly to produce low-carbon electricity. For example, in 2023, around 507 GW of renewable energy capacity was added worldwide, which is nearly a 50% increase compared to 2022. This upward trend is expected to continue. Based on current policies and market trends, global renewable capacity is estimated to reach between 7,300 GW and 8,130 GW by 2028. Grid-enhancing technologies (GETs) have been introduced to reduce power congestion and improve the transmission capability of existing grids. Their role is becoming more important as large-scale, variable renewable energy sources are added to the network. GETs include hardware and software solutions implemented within the current energy infrastructure to increase its capacity, improve flexibility, and boost reliability. These technologies have shown their effectiveness

in power systems globally. They help to reduce congestion, support the connection of renewable sources, cut down on the curtailment of renewable generation, and improve situational visibility and operational flexibility. As electrical grids continue to age, the importance of GETs will grow; however, significant research is still needed to determine when, where, and which specific technologies or combinations should be used[1].

2.LITERATURE REVIEW:

Solar-grid integration refers to the technology and processes that allow large amounts of solar energy from PV or CSP systems to connect to the existing electrical grid. This integration requires careful planning and attention, especially in areas like manufacturing, installation, and operation of solar components. The amount of solar power entering the grid must connect smoothly to the transmission network. Achieving this requires a solid understanding of how solar energy impacts the grid at various connection points. A photovoltaic plant uses PV modules to supply power to the grid and consists of several components. However, the inverter is the key part of the integration process. Other components include the PV array (solar modules), generator junction box (GJB), metering system, grid-connection interface, and both DC and AC wiring. Inverters play a central role in any solar power system and are often seen as the control unit or “brain” of the system. Their main job is to change direct current (DC) from the PV modules into alternating current (AC), which is the type of electricity used by homes and businesses. Inverters must provide stable voltage and frequency, even when the load changes, and they need to manage reactive power for reactive loads. Besides conversion, they ensure that different parts of the system work well together and allow solar electricity to flow into the grid

with maximum efficiency. Thus, the overall performance of a PV installation relies not only on the placement, connection, and quality of the PV modules but also heavily on the reliability and efficiency of the inverter[1]. The physical behavior of a PV cell should be studied early on to find a solution for the partial shading problem. Photocells form the scientific basis of photovoltaic devices. When they receive light, photocells release electrons. The energy related to their movement is described by which represents the energy gained by an electron after it is emitted. This energy is calculated as the product of the electron charge and the voltage applied to the photocell. While a photocell generates current, the number of electrons released directly depends on the intensity of the incoming light. The voltage across the cell greatly affects both the energy of each electron and the total electrical output from the photocell[2]. In a **conventional solar cell**, electrical power **cannot be produced after sunset**. However, under the **anti-solar cell concept**, it becomes possible to generate electricity from solar cells **during nighttime**. In this concept, the **Earth acts as a heat source**, while the **night sky functions as a radiative heat sink**, drawing from principles used in **radiative photovoltaics and advanced radiative cooling technologies**. The **operating physics** of thermo-radiative cells closely resembles that of standard photovoltaic cells. When a **p–n junction** reaches **thermal equilibrium** with its environment in darkness, the **random photon absorption** by the cell is balanced by its **random photon emission**, keeping the **Fermi level constant** throughout the semiconductor[3]

The benefits of solar energy compared to traditional energy sources are discussed. The text starts with a brief history of solar energy, followed by an analysis of the thermal performance of various

solar collectors. These include parabolic troughs, parabolic dishes, Fresnel lenses, solar chimneys, compound parabolic collectors, and flat-plate collectors. Their efficiency is also evaluated. The applications of different solar thermal collector systems are highlighted. These applications include water heating, space heating and cooling, refrigeration, industrial process heating, and thermal power generation. Solar energy technologies can be used in many practical ways. The industrial sector uses about 38% of global energy. This makes it an important indicator of a nation's development. It is essential to diversify energy sources and include renewable options like solar energy. This helps ensure a stable energy supply and reduces emissions from fossil fuel use[4]

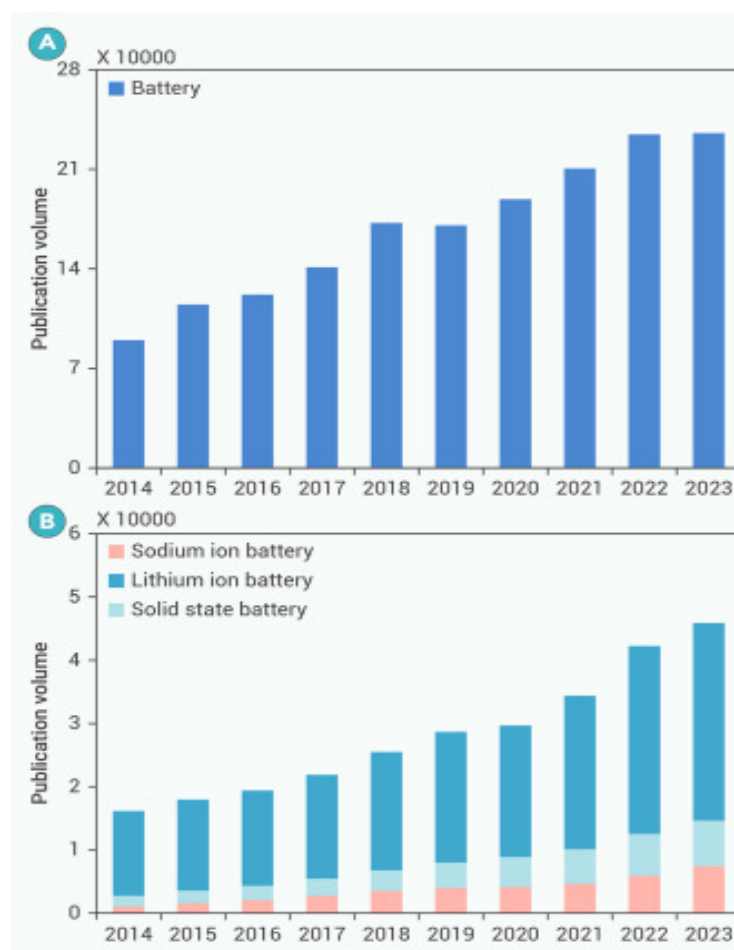


figure 1

The percentage of renewable energy produced globally has soared at the fastest rate ever recorded, and the rise in global power output has largely been attributed to renewable energy sources, with solar and wind generation accounting for the observed increase as can be noted in Figure 1[5]

METHODOLOGY:

The power sector is managed by both the Central and State Governments. At the central level, the Ministry of Energy is responsible for developing and regulating electricity. This includes creating policies for power generation, transmission, distribution, and consumption. The Electricity Supply Act led to the establishment of several bodies, such as the Central Electricity Authority (CEA) and various State Electricity Boards (SEBs). The CEA has several key functions. It develops a national power policy, coordinates and plans the growth of the power sector, evaluates power projects from a technical and economic standpoint, monitors project implementation, ensures that the power system operates efficiently, promotes research and design, assists state governments and electricity boards, collects and publishes power industry statistics, and offers design, consultancy, and training services for engineers. To support coordinated development, Regional Electricity Boards (REBs) were created in 1964 in each of the five regions. Their goal is to achieve integrated operations among the states and build a unified National Grid. The five regions are Northern, Western, Southern, Eastern, and North Eastern, each with its own headquarters and member states. Additionally, Regional and State Load Dispatch Centres (RLDCs and SLDCs) were set up. These centers ensure the best use of generation and transmission systems, maintain the load-frequency balance, and coordinate power flow. They use computerized telemetry and load-control systems to improve efficiency. Despite these efforts, many power projects in the country have experienced significant delays in commissioning. The main reasons for these delays include insufficient funding, a shortage of construction materials, late delivery of equipment, poor coordination among construction teams, labor disputes, and a lack of skilled workers. These issues continue to impact the timely completion and overall effectiveness of power sector projects. It is important to understand the traits and typical uses of nanophotocatalysts before looking into how they work in solar energy rechargeable batteries and other new areas. A photocatalyst is a substance that speeds up a photochemical reaction during photocatalysis. Photocatalysts are mainly divided into two main types based on the physical forms of the photocatalysts and the reactants involved: homogeneous photocatalysts and heterogeneous photocatalysts.[6]. Water splitting includes two important processes: the hydrogen evolution reaction (HER) and the oxygen evolution reaction (OER). The OER is especially hard to improve because it involves several

electron transfers, which leads to slower reaction rates than HER. As a result, making the OER efficient continues to be a significant challenge. Research in photocatalytic water splitting mainly targets two areas: creating photocatalysts with better catalytic performance and developing materials from abundant elements to ensure sustainability and lower costs [6]. Nanophotocatalysts have shown strong results in water splitting and decomposition reactions. They can also be used in fuel cells to generate clean and sustainable electricity. For instance, Cui et al. used TiO_2 nanoparticles as the photoanode to create a solar-powered, storable fuel cell. This cell can directly produce electricity while also breaking down organic pollutants in wastewater. The system generated a steady voltage of about 0.6 V at a constant current of around $20 \mu\text{A cm}^{-2}$, achieving almost total removal of contaminants[Solar energy storage in the rechargeable batteries[6]]. A clear understanding of the properties and traditional uses of nanophotocatalysts is crucial before looking into their role in solar energy rechargeable batteries and other new areas. A photocatalyst is a material that speeds up a photochemical reaction during photocatalysis. Depending on the physical state of the photocatalyst and the photocatalysts are usually divided into two groups: homogeneous photocatalysts and heterogeneous photocatalysts[7]. The percentage of renewable energy produced globally has increased at the fastest rate ever recorded, with the rise in global power output largely attributed to renewable energy sources. In particular, solar and wind generation account for most of this growth, as illustrated in Figure 2[5]

FIGURE 2

RESULT:

Rechargeable batteries have become one of the best technologies for storing electrical energy and are widely used today to meet increasing power needs. Unlike photovoltaic systems, which only turn solar energy into electricity, rechargeable batteries handle both the conversion and storage of energy in electrical and chemical forms. Combining solar energy with high-performance rechargeable batteries can support large-scale energy storage and improve overall efficiency and sustainability [6]. The effectiveness and performance of the battery depend on these factors: 1. Capacity of the battery 2. Efficiency of the battery 3. Depth of discharge Battery capacity refers to the amount of energy that the battery can store. If a 12v battery has a capacity of

500 A.h, the in battery is: $\text{Energy} = \text{Voltage} * \text{Current} * \text{Time} = 12\text{v} * 500 \text{ A.h} = \text{w.h.}$ It is the

ratio of the output energy from the battery to the input energy that the battery needs to charge. If the energy that the battery needs to charge is (6000 wh) and the energy that can be obtained from this battery is (4800 wh), then the efficiency of this battery is: $\text{Battery efficiency} = (\text{output energy}) / (\text{input energy}) * 100\% = (4800 \text{ w.h}) / (6000 \text{ w.h}) * 100\% = 80\%$. A grid-connected photovoltaic (PV) power system is a setup that generates electricity directly linked to the utility grid (energy.gov, n.d.). This system usually includes solar panels, an inverter, and the necessary parts to create a safe and stable connection to the grid. Grid-connected PV systems work well for many uses, such as residential, commercial, and large-scale installations. Unlike off-grid solar power systems, grid-tied systems generally do not need battery storage. When the solar panels produce more electricity than the connected load uses, the extra energy is automatically sent back into the utility grid. This process ensures efficient energy use and lessens reliance on batteries. In residential setups, grid-connected rooftop solar systems usually have a capacity of around 10 kilowatts. This amount is typically enough to meet household energy needs. Any excess electricity generated goes back into the utility grid, allowing other consumers to use this surplus. A dedicated bi-directional meter monitors the transfer of excess power. It tracks both the energy supplied to the grid and the energy consumed from it. Sometimes, the PV system may produce less electricity than the household normally uses. This can happen due to factors such as weather, shading, or system efficiency. In these cases, the household automatically draws the extra power needed from the utility grid, ensuring a steady energy supply.[3]

DISCUSSION:

A quantitative research method was used for this study. Data was collected through a structured questionnaire survey. Collecting primary data was necessary because there was no relevant secondary information on the chosen topic. A convenience sampling technique helped us reach respondents who were easy to access and willing to participate. The questionnaire was sent to 300 professionals working in various sectors of the economy, such as manufacturing, production, supply chains, customer services, construction, energy, and other related fields. We received 105 valid responses, giving us a response rate of 35%. The structured questionnaire had three sections:[3]The first section introduced the topic and outlined the study. The second section included the main survey questions. The final section provided space for respondents to offer feedback and suggestions for improving the study. Participants rated their responses using a 5-point Likert scale, with values from 1 to 5 for each question. The study looks at the current state of solar energy and explores global and national trends, especially in India. Right now, India contributes about 9 GW to global solar energy production, ranking seventh worldwide. Solar energy use in the country can be divided into three main types: government projects, private installations, and individual rooftop systems. Each of these areas has significant untapped potential, showing plenty of chances to increase solar energy adoption across the nation[3]. The study identified the main challenges and barriers that slow down the growth of solar power plants. These issues were evaluated using the Relative Importance Index (RII). The most critical concerns were related to the costs of setting up solar installations. While some government programs provide subsidies for solar panels and storage batteries, these incentives only apply when the equipment is bought from approved agencies. Unfortunately, the prices offered by these agencies are often higher than those from private suppliers and local markets.[4]To fix this issue, the study suggests that subsidy benefits should be available no matter who the vendor is. This would ensure fair access for everyone involved in the solar sector. Overall, the research offers useful insights into the structural and operational challenges that hinder the growth and effective use of solar energy in India.[8]The study was carried out using a structured questionnaire survey and primary data collection. The 105 responses received, while useful for analysis, represent a relatively small sample size and may not be sufficient to generalize the findings on a broader scale. Therefore, it is recommended that similar studies be conducted across various regions of the country to gain a more comprehensive understanding of the challenges and barriers associated with implementing solar energy as an alternative to conventional energy sources in the Indian context[1]

CONCLUSION:

Currently, due to the steady rise in temperatures in recent years, it is important to consider how temperature affects battery performance when choosing the right battery for system design. Temperature is key in determining both the efficiency and safety of solar batteries. As the battery temperature rises, internal chemical reactions speed up, which initially improves the storage capacity. Studies indicate that increasing the temperature from 25 °C to 45 °C can result in a 20% boost in maximum storage capacity. However, this advantage has downsides. Higher temperatures gradually decrease the available capacity and significantly shorten the battery's overall lifespan. Lithium batteries have benefits like being lighter, more compact, and needing less maintenance. However, they come with a higher initial cost and are more sensitive to temperature changes. On the other hand, flooded and gel batteries are commonly used in Iraq because they are more affordable, usually costing just one-fourth to one-sixth of the price of lithium batteries. They also offer an acceptable depth of discharge (DOD) and are better suited for high-temperature conditions, making them a practical choice for the local environment.[2].

REFERENCE:

- [1] T. Su, J. Zhao, A. Gomez-Exposito, Y. Chen, V. Terzija, and J. P. Gentle, “Grid-enhancing technologies for clean energy systems,” *Nat. Rev. Clean Technol.*, vol. 1, no. 1, pp. 16–31, Jan. 2025.
- [2] K. N. Nwaigwe, P. Mutabilwa, and E. Dintwa, “An overview of solar power (PV systems) integration into electricity grids,” *Mater. Sci. Energy Technol.*, vol. 2, no. 3, pp. 629–633, Dec. 2019.
- [3] S. R. G. Weliwaththage and U. S. P. R. Arachchige, *Solar Energy Technology*. Sri Lanka.
- [4] W. Carrión-Chamba, W. Murillo-Torres, and A. Montero-Izquierdo, “A review of the state-of-the-art of solar thermal collectors applied in the industry,” *Ingenius. Revista de Ciencia y Tecnología*, no. 27, pp. 59–73, 2022.
- [5] G. G. Njema, R. B. O. Ouma, and J. K. Kibet, “A review on the recent advances in battery development and energy storage

-
- technologies,” *J. Renew. Energy*, vol. 2024, pp. 1–35, May 2024.
- [6] Z. Wang, R. Liu, J. Wang, B. Wang, M. Zhu, and S. Hu, “High energy conversion efficiency and cycle durability of solar-powered self-sustaining light-assisted rechargeable zinc–air batteries system,” *Energy Storage Mater.*, vol. 74, no. 103897, p. 103897, Jan. 2025.
- [7] R. K. Chougale, Ed., *A Review Paper on Solar Grid Integration in India: Challenges, Strategies and Future Prospects Dr.*
- [8] W. Carrión-Chamba, W. Murillo-Torres, C. Naranjo-Ulloa, K. Valdivieso-García, A. Montero-Izquierdo, and I. Acosta-Pazmiño, “Design and implementation of a web-based residential energy assessment platform: A case study in Cuenca, Ecuador,” in *Congress on Research, Development and Innovation in Renewable Energies*, in Green energy and technology. , Cham: Springer International Publishing, 2022, pp. 121–136.